# Satyajit Ghana

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## Skills

**Languages:** Python, C/C++, JavaScript, TypeScript, Java, Haskell, Golang

**Technologies & Tools:** PyTorch, React, THREE.js, Docker, Deepstream, GStreamer, ROS, SLAM, 3D Reconstruction, Kubernetes, Torchserve, LLamaIndex, LangChain, ISTIO, Blender (BPY), Redis, FastAPI, PCL, Open3D, CloudCompare, Flutter, AWS, Azure, S3, Kinesis, WebRTC, Lambda, Sagemaker, ECS, EKS, Kafka, RabbitMQ, gRPC, Kubeflow, Hydra

## Work Experience

**Inkers Technology Pvt Ltd, Bangalore** April 2023 - Present

***Head of Engineering***

- Led the development of new laser scanning device prototypes, with improved laser accuracy, RGB and Thermal Camera integration.
- Developed a modified SLAM Algorithm that allowed scanning in small and confined areas and reduced drifts by 50%
- Developed a point cloud cleaning algorithm that reduced point cloud thickness from 2cm to 1cm while preserving details.
- ROS1, C++, PCL, Eigen, Python, Open3D, CloudCompare

**The School Of AI, Bangalore** Feb 2022 - Present

***MLOPS Instructor***

- Designed and taught the EMLOV2 and EMLOV3 Course
- Course included, Docker & Docker Compose, PyTorch Lightning & ML Project Setup, Data Version Control, Experiment Tracking, HP Optimization, Gradio, AWS Crash Course, Serving Model w/ FastAPI & Redis, Understanding CLIP, Deploy on Lambda, TorchServe, Model on Browser, Explainability, Kubernetes, EKS, ISTIO, Kiali, KServe, Kubeflow Pipelines, Canary Deployment, Prometheus & Grafana, PEFT, LoRA, Bits&Bytes, vLLM, Dreambooth, Stable Diffusion, SDXL, Fine Tuning LLM, LLM w/ Vision, GGML & LLama.CPP, LLamaIndex & LangChain.
- PyTorch, HuggingFace, Docker, AWS, EC2, S3, ECR, EKS, ECS, EFS, Kubernetes, TorchServe, GGML

**Inkers Technology Pvt Ltd, Bangalore** April 2022 - May 2023

***Deep Learning Software Engineer***

- Optimized the Point Cloud Cleaning and Meshing Algorithm to produce meshes with $\pm 1cm$ deviation.
- Developed a module for Construction Defects Identification and tracking in 3D Textured Mesh, by projecting segmentation to 3d and clustering.
- Designed and Trained custom Transformer Model for 3D mesh part segmentation, which produced material quantities used in construction.
- Optimized mesh sizes from 1.2GB to 120MB using texture optimization, compression and model decimation on Blender, allowing the browser to render these models.
- Migrated the entire Observance WebApp from Bootstrap to TailwindCSS and Prime React, and created an entire Task Management Module with Roles and Authorization.
- Wrote algorithms to synchronize Lidar, Camera and IMU, which improved the 3D SLAM algorithm and reduced drifts.
- Optimized Rendering to "On-Demand" to reduce GPU Load by 30%
- Developed the Floor Plan deviations computation with Hausdorff Distance on the Frontend
- Built a one-of-a-kind thermal camera integration with 3d scanning to produce 3D Thermal Maps, and visualized this on the frontend.
- PCL, ROS, Open3D, Blender, PyTorch, TailwindCSS, React, THREE.js, MeshLab, CloudCompare

**Inkers Technology Pvt Ltd, Bangalore** July 2022 - April 2022

***Associate Deep Learning Software Engineer***

- Developed the fullstack webapp for Observance.AI an end-to-end Construction AI Platform.
- Migrated facial recognition onboarding to AWS Lambda, enabling simultaneous onboarding of people in minutes with up to 2000 parallel Lambdas.
- Built a 3D LiDAR Scan Visualization Platform featuring model annotations, area, length, angle, volume measurements, sync multiple renderers, and with post processing capabilities.
- Optimized the data processing pipeline with HPC instances (p3/p4, g4dn, r6i) and processed 1-2TB of data daily using S3 and spot instances with 128 cores and 1024GB RAM.
- Implemented 3D reconstruction and texturing using SLAM with LIDAR and RGB Camera, and automated the processing pipeline with Blender Scripting for structural defect identification.
- Developed a Camera to Lidar Texture Optimization algorithm in PyTorch3D, which reduced the photometric projection error.
- React, Flutter, DevExtreme, THREE.js, Frappe, 3D Reconstruction, PyTorch, LiDAR, AWS, EC2, S3, ECR, Kinesis Video Streams

**Inkers Technology Pvt Ltd, Bangalore** June 2020 - July 2021

***Visual Processing Intern***

- Developed a driver monitoring system that estimates driver state from sugar levels, pose estimation, and eye landmarks, and created a real-time Flutter app that receives updates over Bluetooth from a Jetson Nano.
- Built a deep learning model that estimates sugar level changes from thermal images.
- Designed a custom GStreamer plugin for DeepStream that performs stereo rectification and additional processing from stereo cameras in near real-time.
- Deployed DeepStream applications on edge devices like Xavier NX using Docker, achieving "stereo" 4K at 30fps for detection and tracking.
- Managed multi-GPU model training on AWS instances with 8x V100, implementing data backup and synchronization to reduce costs.
- Optimized YOLOv4-Tiny for TensorRT, achieving a 2x performance boost, and worked with open-source libraries for TensorRT optimization and model integration with Python and C++.
- Created a synthetic dataset generator that optimizes detection models for robustness against camera glitches, motion blurs, and object sizes, generating 80K images from 100-200 initial images, resulting in a model that detects objects of size 10px x 10px
- Deepstream, Gstreamer, TensorRT, OpenCV, Docker, Flutter, AWS, PyTorch

## Education

**Ramaiah University of Applied Sciences** Aug 2017 - March 2021

B.Tech in Computer Science and Engineering ***CGPA: 9.0/10***

Relevant Coursework: Object Oriented Programming, Databases, Discrete Maths, Data Structures and Algorithms, Operating Systems, Computer Networks, Machine Learning, Data Mining, Advance Data Structures and Algorithms, Information Retrieval, Image Processing